

Ronny Cravioto-Ross

UBC Engineering Physics Student

@ ronny.cravioto@gmail.com c 248nonny.github.io 248nonny r-cravioto-ross Vancouver, Canada

Engineering Physics student with experience writing Rust, C/C++, and Python for normal and bare metal (no_std) targets, experience with robotics and control systems, skilled at learning fast, asking the right questions, and building things from the ground up. Conscious of software design principles (e.g. design by contract, encapsulation, etc), yet passionate for rapid prototyping and getting hands dirty.

Experience and Projects

Incoming C++ Controls Intern – *Kodiak AI - Mountain View, California*

(Future Co-op, On-site) 2026 May. — 2026 Aug.

- Selected for a competitive internship focusing on robust, performant C++ development and control systems.
- Anticipated work includes contributing to a production-level C++ code base subject to formatting standards and stringent code reviews, and physical modelling of vehicle dynamics.

Research Associate – *Blue Sky Spectroscopy - Lethbridge, Alberta, Canada*

(Co-op, On-Site) 2025 Jan. — 2025 Apr.

- Developed multi-channel PID temperature controller through the full design process (PCBs w/ KiCAD, software, testing, etc), for temperature control in large volume cryostats.
- Created and tested PID implementation and I2C ADC driver in C++.
- Reliably controlled temp to within $\pm 3\text{mK}$ of setpoint, 40% better than the target $\pm 5\text{mK}$.
- Worked with vacuum and cryogenic systems, built cryostat test jigs and wire harnesses.
- Worked with a high degree of autonomy, taking full responsibility for designing and implementing major design components.

Robot Summer (UBC Engineering Physics Competition) - First Place of 15 Teams! – *See this video; purple bot is mine: youtu.be/nPeLP5-Fo64*

(UBC Course, Project) 2025 May. — 2025 Aug.

- Worked with 3 peers to develop an autonomous robot from scratch, from mechanical to electrical to software, with resilient design as our core philosophy, on a tight 6 week timeline.
- Won 1st place (of 15) thanks to my team's philosophy of fast iteration and ability to pivot
- Wrote ~50% of firmware codebase (C++, 4.5k lines total), using FreeRTOS and implementing PWM/H-bridge based PID motor control, bit-banging shift registers, I2C sensors, interrupt-based quadrature encoder reader, interrupt-based stepper signal wavegen, odometry position tracking, cubic spline inverse trig functions for arm positioning with constrained compute.
- Designed and built all robot electrical systems (incl. H-bridge and motherboard PCBs).
- Created (in a weekend) a graphical serial debug interface in rust, allowing for live PID tuning, odometry visualization, and arm control. UART (with ring buffers for efficiency) was used, and commands/numeric values (including floats) were sent as raw binary. This became an invaluable insightful resource, allowing us to troubleshoot smarter (not harder).

Neural Network from Scratch (Rust) – *Implemented a two layer feed-forward NN from scratch to detect handwritten numbers.*

(Personal Project) 2025 Feb. — 2025 Feb.

- Used book as guideline for how to construct/train a NN to recognize handwritten numerical digits with the MNIST dataset.
- Implemented multi-threaded stochastic gradient descent with hand coded linear algebra, from scratch (no math/AI/ML deps).
- Created GUI for training and testing with the egui graphics rust crate. After training, user can draw on a canvas, and the NN accurately guesses what the user draws (digits from 0-9) in real time.

Guitar Multi-Effects Pedal with Bare Metal Rust (In Progress)

(Personal Project) 2025 Apr. — Ongoing

- Goal: Create a low-latency guitar effects platform using the RPi Zero 2 W and I2S 24/32 bit 192kHz ADC and DAC.
- In-progress: I2S ADC and DAC breakout PCB has been built and partially tested.
- Completed: Run Linux on one of 4 ARM Cortex A53 cores and execute bare-metal on the other three, allowing Linux drivers to be used alongside bare-metal speed and performance. MMU has also been set up (translation tables, arm registers/assembly).
- Ongoing: Develop communication protocol/driver between Linux userspace and bare-metal effects for real-time parameter tuning (key for 'wah' effect and similar) and signal chain reconfiguration.
- Ongoing: Develop ultra-tight DSP pipeline with the 3 bare metal cores, possibly using custom fixed-point representation, dma, cpu mailboxes, cache prefetching, and SIMD assembly optimizations.

Education

The University of British Columbia - Current Student

B.ASc in Engineering Physics

Minor in Honours Math

Skills/Exposure

Software: Rust, C/C++, Bare-metal, Embedded, Python, Java, Linux + Commandline, Kernel Compilation, Data logging + analysis,

Electrical: KiCAD, Circuit Design, Oscilloscope, Spectrum Analyzer, Soldering,

Mechanical: 3D Printing, Vacuum/Cryo, Solidworks/Fusion360/FreeCAD,

Linux Kernel NVIDIA Workaround

(Personal Project) 📅 2024 Aug. — 2024 Aug.

- Tried to enable hybrid graphics for my Linux running (NixOS) laptop, but graphics card refused to sleep.
- Used kernel logs and ftrace to figure out that a faulty ACPI call caused removal of NVIDIA audio device.
- Wrote a kernel patch to block removal of NVIDIA devices by injecting an if statement to 'acpi_hotplug_work_fn.'
- Idle power went from 14W to 8W, significantly improving battery life.